



# Cambridge (CIE) IGCSE Maths: Extended



Your notes

## Differentiation

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## Differentiation

### What is a gradient function?

- Recall that the **equation of a curve** gives the **y-coordinate** of a point when you substitute in its x-coordinate
  - For example,  $y = x^2 + 3x + 5$ 
    - Substitute  $x = 2$  in to get  $y = 2^2 + 3 \times 2 + 5 = 15$
    - The point  $(2, 15)$  lies on the curve
- A **gradient function** gives the **gradient** of the curve at a point when you substitute in its x-coordinate
  - A gradient function is written as  $\frac{dy}{dx} = \dots$ 
    - pronounced "dy by dx"
    - This refers to the change in y over change in x
  - For example, the gradient function  $\frac{dy}{dx} = 2x + 3$ 
    - Substitute  $x = 2$  in to get  $\frac{dy}{dx} = 2 \times 2 + 3 = 7$
    - The gradient at the point where  $x = 2$  is 7
- The gradient function  $\frac{dy}{dx}$  can also be called the **derivative** or **derived function**

### What is differentiation and how does it work?

- Differentiation** is an **algebraic method** that changes the **equation** of a curve,  $y = \dots$ , into a **gradient function**,  $\frac{dy}{dx} = \dots$
- To differentiate a power of  $x$ , **bring down the power** and **reduce the power by 1**
  - Differentiating  $y = x^5$  gives  $\frac{dy}{dx} = 5x^4$
  - Differentiating  $y = x^8$  gives  $\frac{dy}{dx} = 8x^7$



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- Differentiating  $y = x^n$  gives  $\frac{dy}{dx} = nx^{n-1}$
- Any **number** that is already **in front** of  $x$  is **multiplied** by the power that was brought down

- Differentiating  $y = 2x^5$  gives  $\frac{dy}{dx} = 2 \times 5x^4 = 10x^4$

- Differentiating  $y = \frac{1}{2}x^8$  gives  $\frac{dy}{dx} = \frac{1}{2} \times 8x^7 = 4x^7$

- Differentiating  $y = kx^n$  gives  $\frac{dy}{dx} = knx^{n-1}$

- Be careful with **two special cases**:

- Differentiating  $y = kx$  gives  $\frac{dy}{dx} = k$

- The  **$x$  disappears** leaving just a number

- For example,  $y = 2x$  differentiates to  $\frac{dy}{dx} = 2$

- This makes sense, the gradient of the straight line  $y = 2x$  is 2

- Differentiating just a number (a **constant** term),  $y = c$ , gives  $\frac{dy}{dx} = 0$

- **Numbers disappear!**

- For example,  $y = 4$  differentiates to  $\frac{dy}{dx} = 0$

- This makes sense, the gradient of the horizontal line  $y = 4$  is zero

USING DIFFERENTIATION TO FIND THE DERIVED FUNCTION

IF  $y = kx^n$

THEN  $\frac{dy}{dx} = knx^{n-1}$

STEP 1: MULTIPLY  
BY THE POWER

STEP 2: TAKE ONE  
OFF THE POWER

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## How do I differentiate sums and differences of terms?



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- The equation of a curve may include a number of different terms
  - You can differentiate each term **individually**
    - Differentiating  $y = x^5 + x^8$  gives  $\frac{dy}{dx} = 5x^4 + 8x^7$
    - Differentiating  $y = 4x^3 - 10x^6$  gives  $\frac{dy}{dx} = 12x^2 - 60x^5$
- Remember the two special cases of:
  - $kx$  differentiating to just  $k$
  - and **constant** terms,  $c$ , differentiating to **zero**
    - E.g. differentiating  $y = 8x + 1$  gives  $\frac{dy}{dx} = 8$

$$y = 3x^2 - 5x + 3$$
$$\frac{dy}{dx} = 6x - 5$$

THE DERIVATIVE OF 3 (A CONSTANT) IS ZERO

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## How do I find the gradient of a curve using the gradient function?

- Find the **x-coordinate** of the point on the curve you're interested in
- Use **differentiation** to turn the equation of the curve,  $y = \dots$ , into the gradient function,  $\frac{dy}{dx} = \dots$
- **Substitute** the x-coordinate into the gradient function to find the gradient
  - The y-coordinate is not needed



Your notes

e.g. FIND THE GRADIENT OF THE CURVE  $y = 3x^2 - 2x$   
AT THE POINT  $(2, 8)$

$$\frac{dy}{dx} = 6x - 2 \leftarrow \text{FIND THE DERIVED FUNCTION}$$

$$\text{WHEN } x = 2, \frac{dy}{dx} = 6 \times 2 - 2 \leftarrow \text{SUBSTITUTE IN THE } x\text{-COORDINATE}$$

$$\frac{dy}{dx} = 10$$

THE GRADIENT AT THE POINT  $(2, 8)$  IS 10

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- If instead you are **given a gradient** and asked to find the **x-coordinate**

- set  $\frac{dy}{dx}$  equal to that gradient and **solve** the equation



### Examiner Tips and Tricks

Don't forget to write the left-hand sides of  $y = \dots$  and  $\frac{dy}{dx} = \dots$  to avoid mixing up the curve equation with the gradient function!

## How is the gradient function related to drawing tangents?

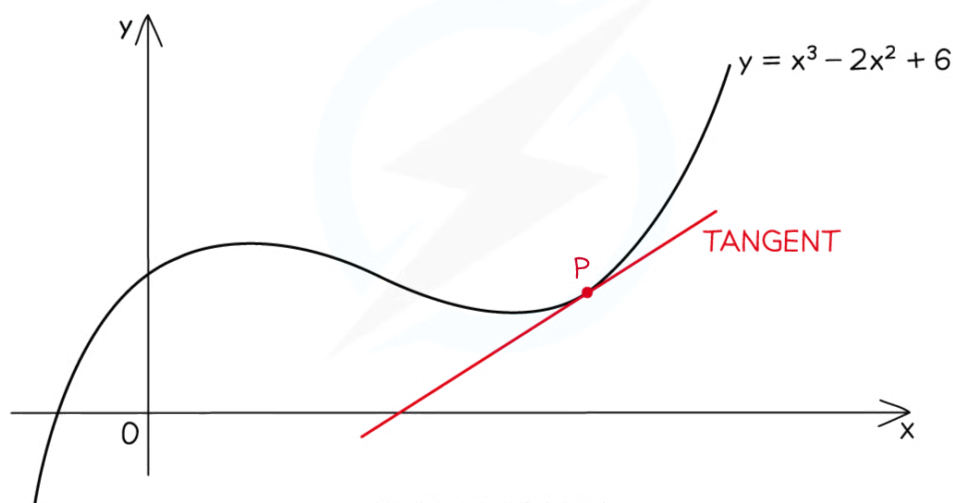
- The gradient of a curve changes as you move along the curve
  - To find the gradient at a particular point you can **draw a tangent** and find its **gradient**
  - This is a **graphical** method that is **not accurate**
    - It depends on how well you draw the tangent
- Instead, you can use **differentiation** to find the **gradient function**, then **substitute the x-coordinate** of the point into the gradient function to find the gradient
  - This is an **algebraic** method that is **exact**
- For example, to find the **gradient** of the curve  $y = x^3 - 2x^2 + 6$  at the point  $P$  where  $x = 2$ 
  - either try to draw a tangent at  $x = 2$  and measure its gradient (see below)

- or differentiate the equation to get  $\frac{dy}{dx} = 3x^2 - 4x$

- Substitute in  $x = 2$  to get  $\frac{dy}{dx} = 3 \times 2^2 - 4 \times 2 = 4$

- The gradient is 4

THE GRADIENT OF THE CURVE AT POINT P  
IS EQUAL TO THE GRADIENT OF THE  
TANGENT TO THE CURVE AT POINT P



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### Worked Example

A curve has the equation  $y = x^3 - 3x^2 - 4x + 1$ .

(a) Find the gradient of the curve at the point  $(1, -5)$ .

Find the gradient function using differentiation

$$\frac{dy}{dx} = 3x^2 - 3 \times 2x - 4 + 0$$

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Substitute  $x = 1$  into the gradient function



Your notes



Your notes

$$\begin{aligned}\frac{dy}{dx} &= 3 \times 1^2 - 6 \times 1 - 4 \\ &= -7\end{aligned}$$

The gradient when  $x = 1$  is  $-7$

(The  $y$ -coordinate of the point is not needed)

The gradient at  $(1, -5)$  is  $-7$

(b) Find the coordinates of the two points on the curve with a gradient of 5.

This is saying that  $\frac{dy}{dx} = 5$

(This is not the same as substituting in  $x = 5$ )

Form an equation using  $\frac{dy}{dx}$  from above

$$3x^2 - 6x - 4 = 5$$

This is a quadratic equation

Bring the terms to one side and solve (for example, by factorisation)

$$\begin{aligned}3x^2 - 6x - 9 &= 0 \\ x^2 - 2x - 3 &= 0 \\ (x - 3)(x + 1) &= 0 \\ x &= 3 \text{ or } x = -1\end{aligned}$$

These are the  $x$ -coordinates of the two points on the curve with gradient 5

Substitute  $x = 3$  into  $y = x^3 - 3x^2 - 4x + 1$  to find the  $y$ -coordinate of this point

$$\begin{aligned}y &= 3^3 - 3 \times 3^2 - 4 \times 3 + 1 \\ &= -11\end{aligned}$$

Similarly, substitute  $x = -1$  into the equation for  $y$

$$\begin{aligned}y &= (-1)^3 - 3 \times (-1)^2 - 4 \times (-1) + 1 \\ &= 1\end{aligned}$$

Write out the two sets of coordinates

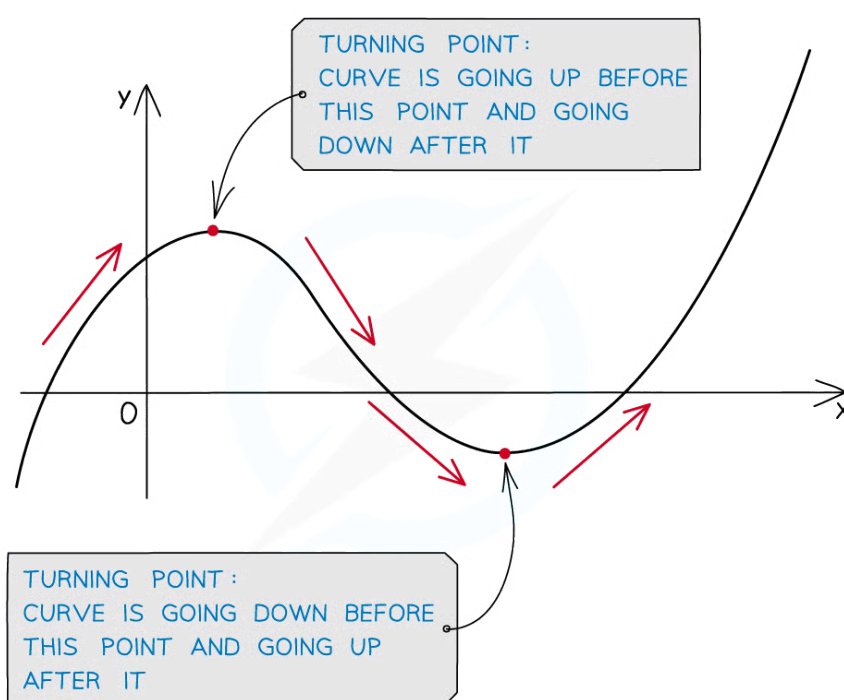
The points  $(3, -11)$  and  $(-1, 1)$  have a gradient of 5



# Finding Stationary Points & Turning Points

## What is a stationary point?

- A **stationary point** is a point on the graph at which the **gradient is zero** (a tangent drawn at this point will be **horizontal**)
  - These *include* peaks (**maximum points**) and troughs (**minimum points**)
- Maximum points and minimum points are collectively known as **turning points**
  - At a turning point a curve **changes** from **moving upwards** to **moving downwards**, or vice versa



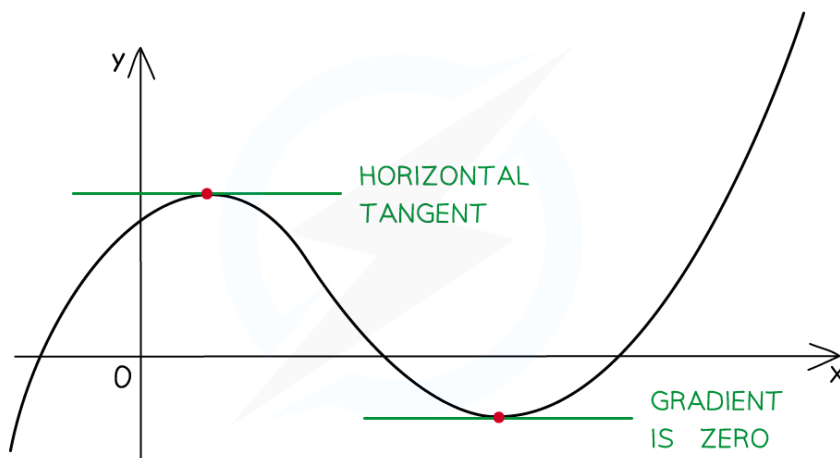
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- At a **turning point** the **gradient** of the curve is **zero**
  - If a **tangent** is drawn at a turning point it will be a **horizontal line**
  - Horizontal lines have a gradient of zero
- Because the gradient at a turning point is zero
  - Substituting the x-coordinate of a turning point into the **gradient function**  $\frac{dy}{dx}$  (the **derivative**) will give an **output of zero**





Your notes



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## How do I find the coordinates of a turning point?

### STEP 1

Find the **gradient function** (derivative)  $\frac{dy}{dx}$  of the **original equation**

- E.g. Find the coordinates of the turning point of the equation  $y = 4x^2 - 8x + 9$

- $\frac{dy}{dx} = 8x - 8$

### STEP 2

Set the **gradient function** (derivative) **equal to zero** and **solve for X**

- This will find the **x-coordinate** of the turning point
- E.g.  $0 = 8x - 8$ , so  $x = 1$

### STEP 3

**Substitute** the x-coordinate into the original equation of the **graph**

- This will find the **y-coordinate** of the turning point
- E.g.  $y = 4(1)^2 - 8(1) + 9 = 5$

### STEP 4

Write out the **coordinates** of the turning point

- E.g. The turning point is at  $(1, 5)$



### Examiner Tips and Tricks

- Remember to read the questions carefully
  - Sometimes only the x-coordinate of a turning point is required



Your notes



### Worked Example

Find the coordinates of the turning point on the curve with equation

$$y = 2x^2 + 8x - 9.$$

Use differentiation to find the gradient function (derivative) of the equation

$$\frac{dy}{dx} = 4x + 8$$

Set the derivative equal to 0 and solve for x

$$\begin{aligned} 0 &= 4x + 8 \\ -8 &= 4x \\ -2 &= x \end{aligned}$$

Substitute the x-coordinate into the origination equation of the curve and solve for y

$$\begin{aligned} y &= 2(-2)^2 + 8(-2) - 9 \\ y &= 8 - 16 - 9 \\ y &= -17 \end{aligned}$$

Form a set of coordinates

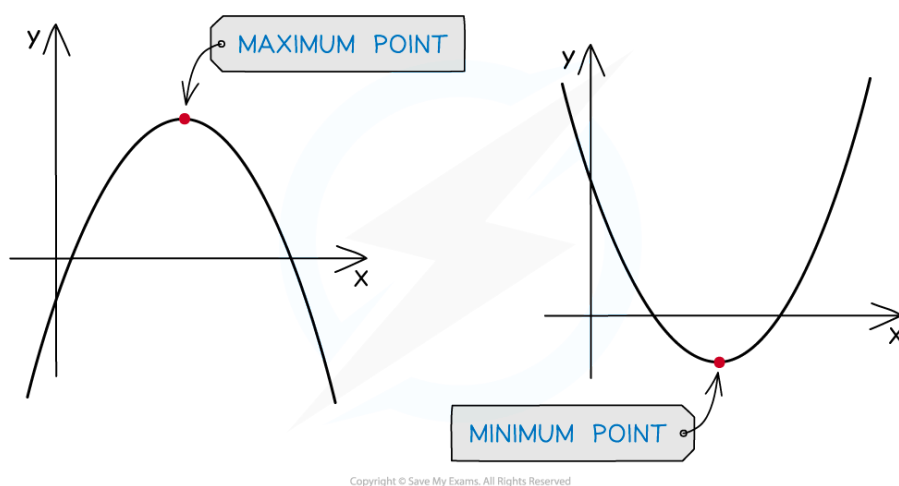
The turning point has coordinates  **$(-2, -17)$**



# Classifying Stationary Points

## What are the different types of stationary points?

- You need to know **two** different types of stationary points:
  - **Maximum points** (this is where the graph reaches a “**peak**”)
  - **Minimum points** (this is where the graph reaches a “**trough**”)
    - These are also called **turning points**
    - Deciding which is which is called **classifying** them (or finding the **nature** of them)



- If a graph has multiple turning points, the one you are interested in can be described as a **local** maximum or minimum point
  - Other parts of the graph may still reach higher/lower values elsewhere

## How do I classify turning points using graphs?

- You can use the **shape of common graphs** to classify different turning points
- To **classify the turning point** on a **quadratic curve** (parabola), remember that:
  - A **positive** quadratic (**positive  $x^2$**  term) must have a **minimum** point
  - A **negative** quadratic (**negative  $x^2$**  term) must have a **maximum** point

## PARABOLAS (QUADRATICS)



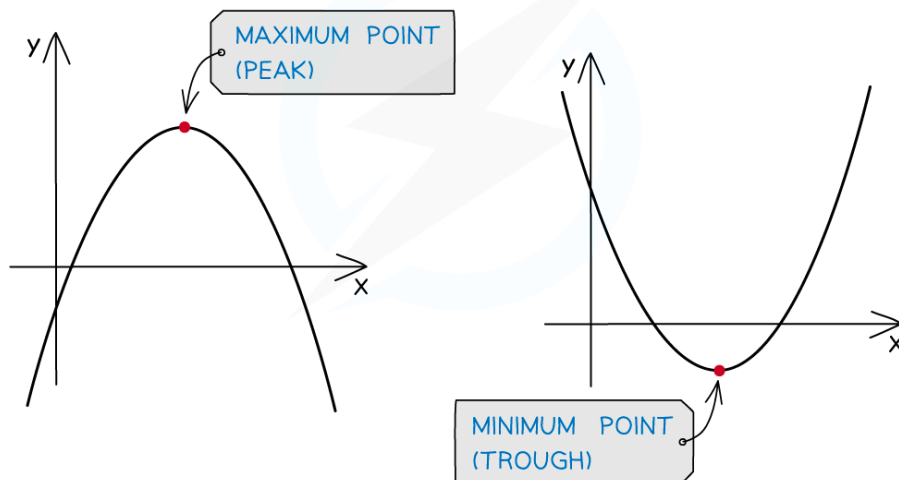
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$$y = ax^2 + bx + c$$

$a < 0$  NEGATIVE PARABOLA

$$y = ax^2 + bx + c$$

$a > 0$  POSITIVE PARABOLA



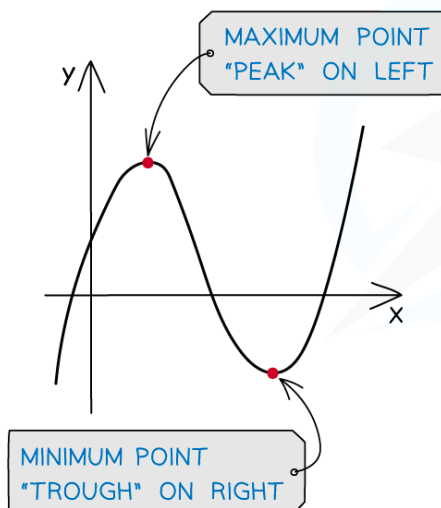
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- To **classify two different turning points** on a **cubic curve**, remember that:
  - A **positive** cubic has a **maximum** point on the **left** and a **minimum** on the **right**
  - A **negative** cubic has a **minimum** point on the **left** and a **maximum** on the **right**

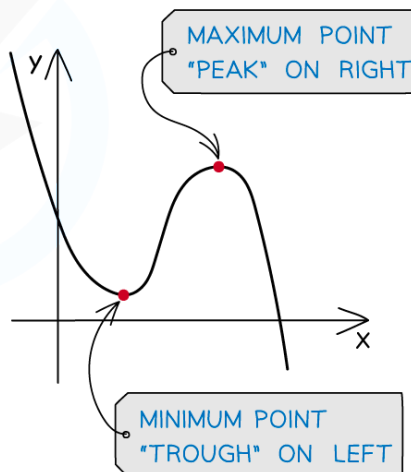


## CUBIC GRAPHS

$$y = ax^3 + bx^2 + cx + d$$
$$a > 0 \text{ POSITIVE CUBIC}$$



$$y = ax^3 + bx^2 + cx + d$$
$$a < 0 \text{ NEGATIVE CUBIC}$$



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## How do I classify turning points using the gradient function (derivative)?

- An alternative method if no graph is available is to use the **gradient function** (derivative)  
 $\frac{dy}{dx}$ 
  - By looking at the **gradient just before and just after** a turning point, you can determine if it is a maximum or a minimum
  - Find the gradient just before and just after a turning point by substituting values near to the value of  $x$  into the gradient function
  - It helps to imagine what the **tangents** to the curve would look like

| Gradient before turning point | Gradient at turning point | Gradient after turning point | Type of turning point |
|-------------------------------|---------------------------|------------------------------|-----------------------|
| Positive<br>/                 | Zero<br>-                 | Negative<br>\                | Maximum<br>/-\        |
| Negative<br>\                 | Zero<br>-                 | Positive<br>/                | Minimum<br>\-/        |

## How do I classify turning points using the second derivative?



Your notes

- Another method is to use the **second derivative**,  $\frac{d^2y}{dx^2}$ , which is the **derivative-of-the-derivative**
  - $\frac{d^2y}{dx^2}$  is pronounced "d-two-y by dx-squared"
- Differentiate the equation once to get  $\frac{dy}{dx}$  then again to get  $\frac{d^2y}{dx^2}$ 
  - This is the same as **differentiating** the original equation for **y twice**
- If the stationary point is at  $x = a$ , substitute  $x = a$  into the expression for  $\frac{d^2y}{dx^2}$  to get a numerical value
  - If the result is **negative**,  $\frac{d^2y}{dx^2} < 0$ , then the stationary point is a **maximum** point
    - This is often the opposite of what people expect!
  - If the result is **positive**,  $\frac{d^2y}{dx^2} > 0$ , then the stationary point is a **minimum** point
  - If the value is **zero**,  $\frac{d^2y}{dx^2} = 0$ , then the test has **failed**
    - You will need to try another method to classify the turning point



### Worked Example

The turning points on the curve with equation  $y = 2x^3 + 3x^2 - 12x + 1$  are  $(-2, 21)$  and  $(1, -6)$ .

Determine the nature of these turning points.

*Method 1: Using the graph*

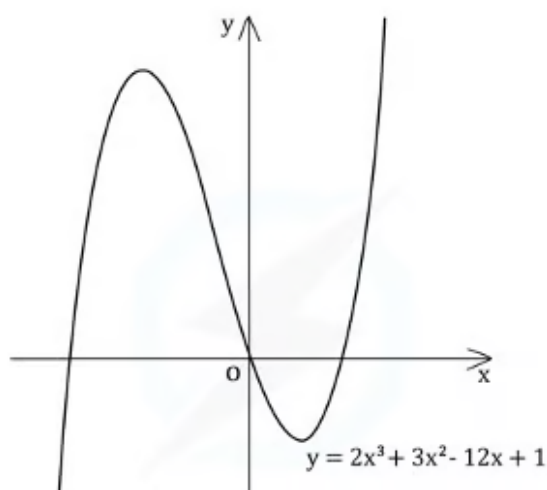
Make a quick sketch of the curve

It is a positive cubic so you know that it will start at the bottom left, have two turning points and go towards the top right

You know the coordinates of the turning points and the y-intercept:  $(0, 1)$   
(the sketch does not have to be perfect!)



Your notes



From the graph, which is a positive cubic,  
(-2, 21) is to the left of (1, -6)

Therefore, (-2, 21) is a maximum point and (1, -6) is a minimum point

Method 2: Differentiation (1<sup>st</sup> derivative)

Differentiate the original equation to find the 1<sup>st</sup> derivative, the gradient function,  $\frac{dy}{dx}$

$$2x^3 \rightarrow 6x^2$$

$$3x^2 \rightarrow 6x$$

$$-12x \rightarrow -12$$

$$1 \rightarrow 0$$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

Find the gradient just before and just after each turning point by substituting a suitable x-value into the gradient function

| Turning point | Gradient before             | Gradient after               |
|---------------|-----------------------------|------------------------------|
| (-2, 21)      | sub $x = -3$                | sub $x = -1$                 |
|               | $6(-3)^2 + 6(-3) - 12 = 24$ | $6(-1)^2 + 6(-1) - 12 = -12$ |
| (1, -6)       | sub $x = 0$                 | sub $x = 2$                  |
|               | $6(0)^2 + 6(0) - 12 = -12$  | $6(2)^2 + 6(2) - 12 = 24$    |

The gradient just before (-2, 21) is positive and the gradient just after is negative,  
therefore the turning point here is a maximum point

The gradient just before (1, -6) is negative and the gradient just after is positive, therefore the turning point here is a minimum point



Your notes

Method 3: Differentiation (2<sup>nd</sup> derivative)

Differentiate the original equation to find the 1<sup>st</sup> derivative

$$2x^3 \rightarrow 6x^2$$

$$3x^2 \rightarrow 6x$$

$$-12x \rightarrow -12$$

$$1 \rightarrow 0$$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

Differentiate again to find the 2<sup>nd</sup> derivative

$$6x^2 \rightarrow 12x$$

$$6x \rightarrow 6$$

$$-12 \rightarrow 0$$

$$\frac{dy}{dx} = 12x + 6$$

Find the nature of each turning point by substituting the value of its x-coordinate into the 2<sup>nd</sup> derivative

A negative result indicates a maximum point (or positive for a minimum point)

$$\text{For } (-2, 21): \frac{dy}{dx} = 12(-2) + 6 = -18$$

The second derivative is negative, therefore the turning point at (-2, 21) is a maximum point

$$\text{For } (1, -6): \frac{dy}{dx} = 12(1) + 6 = 18$$

The second derivative is positive, therefore the turning point at (1, -6) is a minimum point





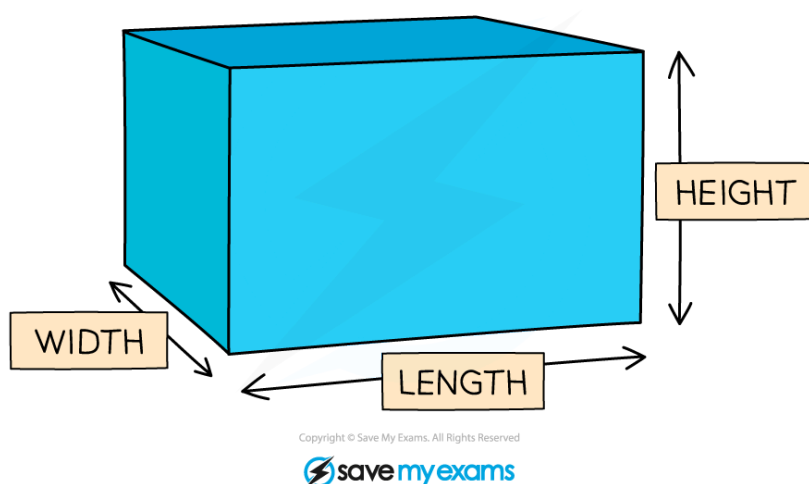
## Problem Solving with Differentiation

### What is problem solving using differentiation?

- You can use the same method of **differentiating curves** to find **turning points** to help with problems involving finding the **maximum or minimum value** of a quantity
  - These are called **optimisation** problems
- Questions may use **different variables**
  - For example,  $A = t^3 - 12t$  and  $\frac{dA}{dt} = 0$

### How do I apply differentiation to different contexts?

- This is easiest explained using an example



- Find the **maximum volume**,  $V \text{ m}^3$ , of the cuboid shown with width 2 m, length  $x$  m and height  $(10 - x)$  m
  - Find a **formula** for its volume in terms of  $x$
  - Volume = width  $\times$  length  $\times$  height
    - $V = 2x(10 - x)$
  - Expand** this into individual terms in  $x$ 
    - $V = 20x - 2x^2$
    - This is a negative quadratic in  $x$  (it has an  $\cap$  shape)



Your notes

- The graph of  $V$  against  $x$  will have a **maximum point** when  $\frac{dV}{dx} = 0$ 
  - Find  $\frac{dV}{dx}$  by differentiating each term
    - $\frac{dV}{dx} = 20 - 4x$
  - Set this **equal to zero** and solve
    - $20 - 4x = 0$  so  $4x = 20$  giving  $x = 5$
    - This is the value of  $x$  at the maximum point (not the value of  $V$ )
  - **Substitute** this value of  $x$  back into the equation for  $V$ 
    - $V = 20 \times 5 - 2 \times 5^2 = 50$
    - The **maximum volume** is  $50 \text{ m}^3$



### Examiner Tips and Tricks

A common problem in the exam is to forget to substitute the value of  $x$  back into the formula!

## How do I know if I have found a maximum value or a minimum value?

- Sometimes there are **two values** of  $x$  for different **turning points** and you need one of them
  - either substitute them both back into the formula
    - See which gives the max value and which gives the min value
  - or use any acceptable techniques for **classifying turning points**
    - For example, using a sketch to see if its a max or min
- Remember that
  - **Positive quadratics** have **minimum** points
  - **Negative quadratics** have **maximum** points

## How do I use differentiation if there are lots of variables?

- Sometimes a formula has lots of letters



Your notes

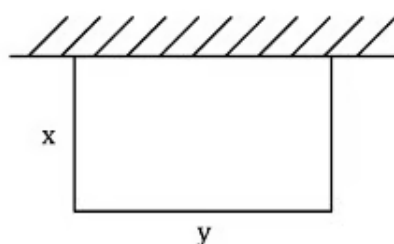
- You need to **find an extra relationship** between these letters
- then **substitute** it into the formula
- If, in the above example, you had width 2 m, length  $x$  m and height  $y$  m then
  - $V = 2xy$ 
    - But you need a formula in  $x$  only
  - You will be **given** an extra piece of information, such as the length and height sum to 10 m
    - Therefore  $x + y = 10$
    - Make  $y$  the subject,  $y = 10 - x$
    - Substitute it into  $V$  to get  $V = 2x(10 - x)$



### Worked Example

A farmer has 60 metres of fencing and wants to fence off the biggest rectangular area possible next to an existing wall.

The area has dimensions  $x$  metres by  $y$  metres, as shown.



(a) Explain why  $2x + y = 60$ .

The wall is not part of the fencing

Find the total length of the three sides of the fence shown

$$x + y + x$$

This must equal 60 metres

The length of the fence must equal 60 metres so  $2x + y = 60$

(b) Show that the area,  $A \text{ m}^2$ , is given by  $A = 60x - 2x^2$ .

Find the area of the rectangle shown

$$A = xy$$

You need a right-hand side in terms of  $x$  only

Make  $y$  the subject of part (a)

$$y = 60 - 2x$$

Substitute this into the area formula

$$A = x(60 - 2x)$$

This is now all in terms of  $x$

Expand

$$A = 60x - 2x^2$$

(c) Find the maximum possible area.

To find a maximum point, set  $\frac{dA}{dx} = 0$

First find  $\frac{dA}{dx}$  by differentiating each term

$$\frac{dA}{dx} = 60 - 4x$$

Set this equal to zero and solve

$$60 - 4x = 0$$

$$4x = 60$$

$$x = \frac{60}{4}$$

$$x = 15$$

This is the value of  $x$  at the maximum point (but not the maximum of  $A$ )

Substitute this value of  $x$  back into  $A$

$$\begin{aligned} A &= 60 \times 15 - 2 \times 15^2 \\ &= 450 \end{aligned}$$

The maximum area is  $450 \text{ m}^2$

(d) Explain how you know that the answer in part (c) is a maximum area, not a minimum area.

See how the area  $A$  depends on  $x$



Your notes

Think about what it would look like as a graph

$A = 60x - 2x^2$  is a negative quadratic curve

A negative quadratic curve has an  $\cap$  shape

There is only one turning point on this graph and it is a maximum point

The curve  $A = 60x - 2x^2$  is a negative quadratic curve

This means it can only have a maximum point, not a minimum point



Your notes